

Dr. Nico Winkel

Astrophysicist · Data Scientist

Mathematical Optimisation · Energy Systems · Industrial Operations

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PROFILE

I am a data scientist building decision systems for energy and industry. At OmegaLambdaTec I lead technical delivery for grid operators and manufacturers, working on mathematical optimisation and energy-system simulation. My work covers the full path from a customer's problem statement to a running system: framing the problem, choosing a method that stays tractable at real scale, and integrating the result into the customer's existing software landscape. My PhD at the Max Planck Institute for Astronomy taught me to own a method end to end and to explain it clearly to experts and non-experts alike.

INDUSTRY EXPERIENCE

07/2025 – now

Data Scientist

Garching, Germany

OmegaLambdaTec GmbH

KI-HEAP4.0 Project Manager & Developer

BayVFP-funded 5-partner consortium with GROB and MEILLER; team of three plus a junior colleague

- Led the team building a CP-SAT scheduler with a constraint formulation new to the FJSSP literature: up to 15% shorter makespan, with multi-week horizons re-planned within minutes. A consortium partner benchmarks its reinforcement-learning scheduler against our provably optimal schedules, while our team runs the performance testing.
- Led the design of the path from optimiser to shop floor (machine event streams, SAP/MES interfaces, a monitor that flags when execution drifts from the plan). A failure module calibrated on historical machine data tests schedules against realistic breakdowns.

Redispatch 2.0 Lead Developer

Customer project for TransnetBW

- Led the implementation of a MILP that clears transmission-grid congestion by re-dispatching continuous, step-discrete and storage assets under ramp, activation and switching constraints.
- Kept the full horizon tractable across hundreds of plants and 96 timesteps by replacing repeated AC power flow with PTDF-linearised branch flows, in a solver-agnostic Pyomo backend with explicit MIP-gap.

RLMate4WP Developer

BMWE-funded research project with CAE Würzburg; AI-based heat-pump control

- Implemented and trained RL agents, compared them against model predictive control and threshold control for heat pumps with PV and battery, validated on a physical building model.
- Model predictive control cut annual energy costs by up to 30% and energy use by almost 20% against standard control under dynamic tariffs.

SKILLS

Methods	MILP and constraint programming · scheduling (FJSSP) · model predictive control · reinforcement learning (PPO, SAC) · statistical inference · time series analysis · discrete-event simulation · simulation-based evaluation
Modelling & solvers	Pyomo, OR-Tools CP-SAT · HiGHS, SCIP, CBC, GLPK
Engineering	Python (NumPy, Pandas, scikit-learn, SciPy, SimPy), SQL, Bash, Git, Docker, HPC · SAP/MES integration · Matplotlib, Plotly, Dash
Domain	Transmission-grid congestion management (Redispatch 2.0) · heat-pump control and building energy flexibility · industrial production planning
Languages	German (native) · English (C1) · Spanish (A2, learning) · French (A1)

MENTORING & TEACHING

09/2025 – 07/2026	Supervisor for Intern Developer OmegaLambdaTec GmbH	Garching, Germany
10/2020 – 09/2021	Tutorial Lecturer Physics & Astronomy University of Heidelberg	Heidelberg, Germany
02/2018 – 11/2019	Private Tutor Mathematics Studentenring Heidelberg	Heidelberg, Germany
04/2017 – 09/2019	Teaching Assistant Physics University of Heidelberg	Heidelberg, Germany

RESEARCH EXPERIENCE

02/2023 – 04/2026	Referee for Scientific Journals Astronomy & Astrophysics · MNRAS · The Astrophysical Journal	
11/2024 – 02/2025	Postdoctoral Researcher, Astrophysics Max Planck Institute for Astronomy Principal Investigator of a 212 h ($\simeq 1.2$ M€) observing programme with ESO VLT: proposal, strategy and acquisition.	Heidelberg, Germany
10/2020 – 10/2024	Doctoral Researcher, Astrophysics Max Planck Institute for Astronomy - Led data reduction and modelling of TB-scale <i>Hubble</i>, <i>ALMA</i> and <i>JWST</i> datasets; built calibration and inference pipelines in Python for instruments in active use. - Coordinated international collaborations resulting in 16 peer-reviewed publications.	Heidelberg, Germany

EDUCATION

10/2020 – 10/2024	Dr. rer. nat. Astrophysics University of Heidelberg · magna cum laude	Heidelberg, Germany
10/2018 – 09/2020	Master of Science in Physics University of Heidelberg · Grade 1.1	Heidelberg, Germany
10/2014 – 09/2018	Bachelor of Science in Physics University of Heidelberg · Grade 2.2	Heidelberg, Germany